PROBLEM STATEMENT

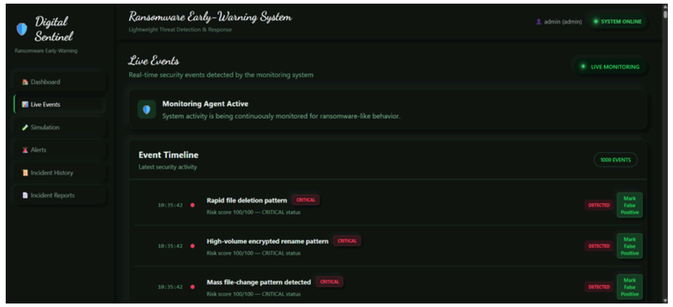
CIPHERVEIL: RANSOMWARE EARLY-WARNING SYSTEM

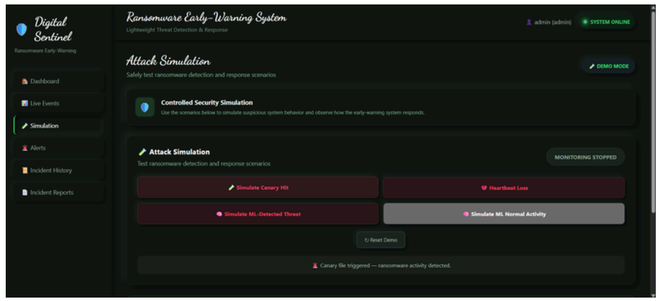
AI/ML-Based Behavioral Monitoring for Ransomware Detection

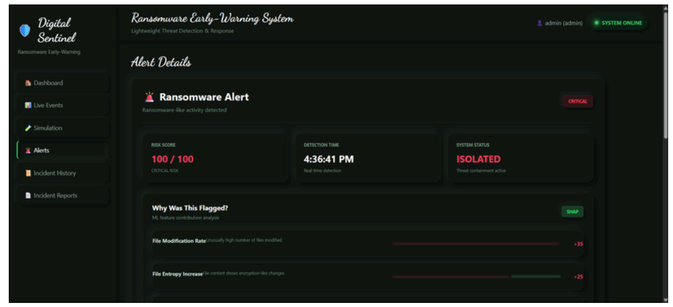
Ransomware attacks can rapidly encrypt files, alter system settings, and spread across connected devices. Without continuous behavioral monitoring and early risk detection, suspicious activity can be difficult to identify and contain before data damage occurs.

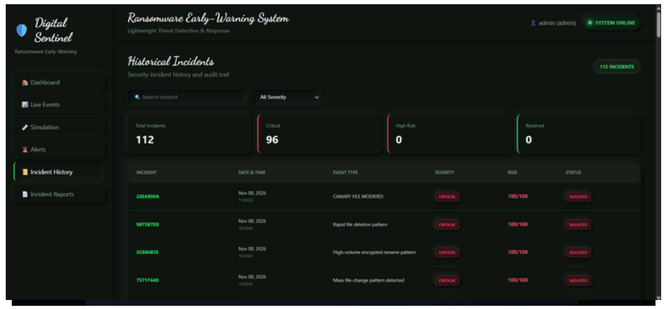
Theme: Cybersecurity, Digital Safety, and Data Protection

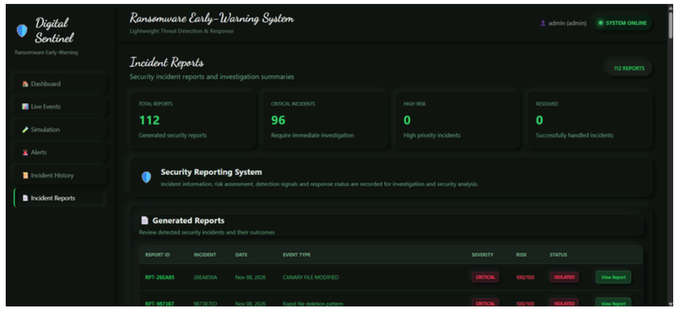


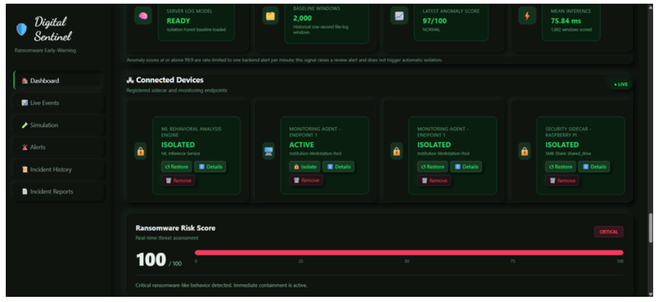


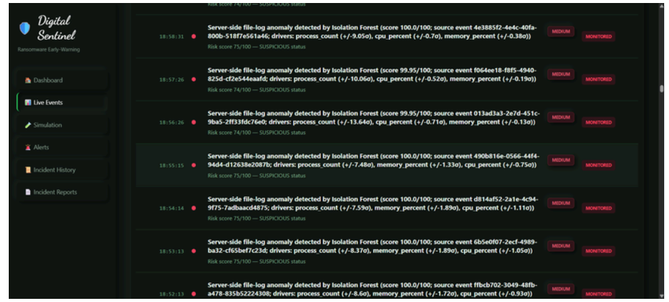












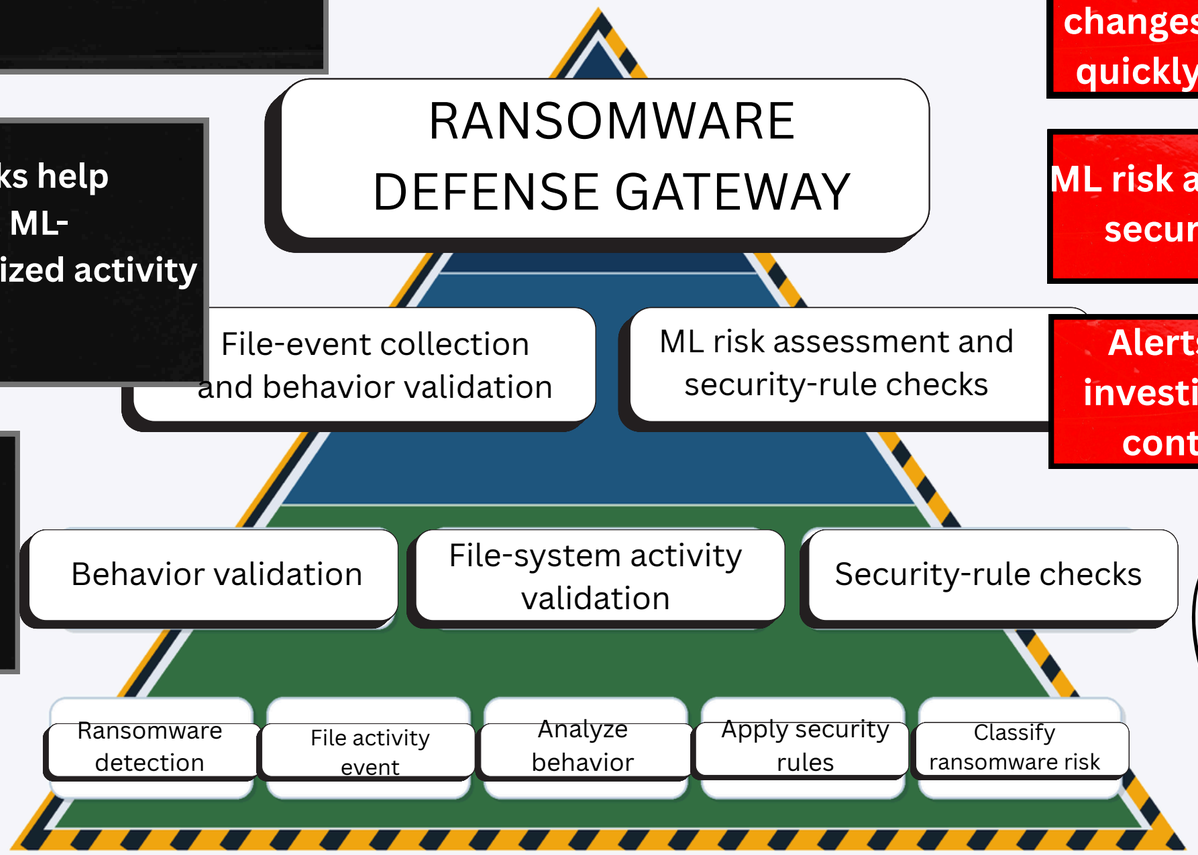
Cipherveil Ransomware Early-Warning Gateway

Monitor device activity, inspect file events, enforce response rules, and detect ransomware

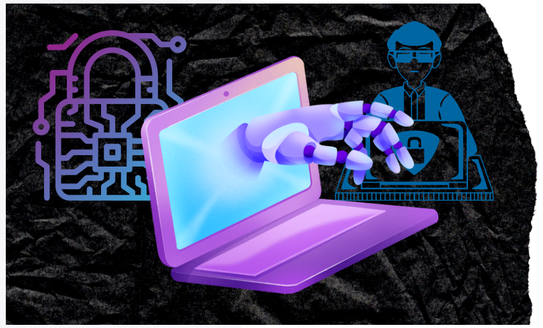
The challenge: Ransomware can encrypt, modify, or delete files rapidly, making suspicious activity difficult to identify in time

Why it matters: Early, explainable checks help teams identify suspicious file behavior, ML-generated risk decisions, and unauthorized activity quickly

CIPHERVEIL: monitors device events, analyzes behavioral patterns, applies security rules, and records alerts for investigation



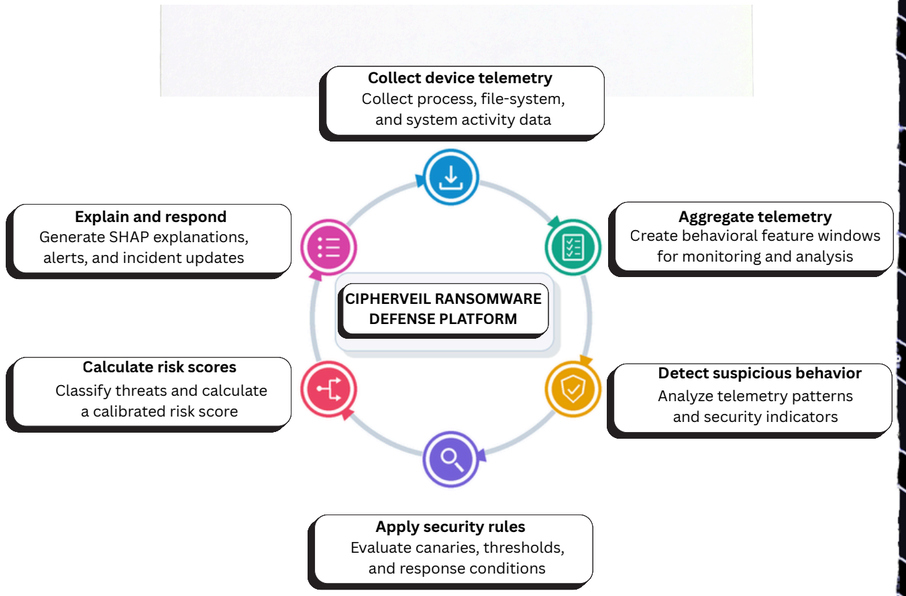
Unmonitored file activity	vs	Cipherveil defense platform
Encryption-like changes occur too quickly to detect		Device activity continuously monitored
ML risk analysis and security rules		Explainable security decisions
Alerts support investigation and containment		Events, alerts, and unified risk dashboard



MONITOR — ANALYZE — DETECT — RESPOND

CIPHERVEIL TECHNICAL APPROACH

END-TO-END THREAT DETECTION WORKFLOW



GITHUB REPO

Detailed Report

CIPHERVEIL	What It Does	Security Purpose
Monitoring Module	Captures process, file-system, and device activity	Generates telemetry windows every second
ML Risk Engine	Runs XGBoost and Isolation Forest models	Produces calibrated threat scores and explanations
Security Rules Module	Evaluates canary checks, thresholds, and response conditions	Creates alerts for suspicious ransomware behavior
Monitoring Telemetry	Stores event windows and monitoring history	Provides device telemetry and anomaly-status summaries
Backend and Dashboard Workflow	Stores events, alerts, incidents, and incident timelines	Visualizes risk scores, explanations, and operator responses

PROCESS DETAILS

Pipeline: Collect, Aggregate, Analyze, Score, Alert

Detect: ML models and security rules calculate a ransomware risk score

Respond: Publish alerts and start incident-response or containment workflows

Explain: Show risk factors, matched indicators, and SHAP-based explanations

TECHNOLOGIES

Frontend
React + Vite

Backend
Python + FastAPI

ML
XGBoost, Isolation Forest, and SHAP explainability

Database
SQLite + SQLAlchemy

Security
Device API keys, canary checks, response rules, and security event logging

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

Prototype stack: Python, FastAPI, React/Vite, XGBoost, and SHAP for telemetry collection, threat classification, explainability, alerts, and dashboard monitoring.

Deployment: The monitoring agent, ML service, backend API, security sidecar, and dashboard run locally as a prototype. Production deployment, distributed endpoint support, scaling, and managed operations require further evaluation.

Cost assessment: The prototype uses open-source technologies and local services, reducing initial implementati on costs. Cloud hosting, maintenance, storage, endpoint management, and operational support will require cost evaluation.

Explainability: TreeSHAP identifies the behavioral features contributing to an ML prediction. The backend records risk scores, detection decisions, contributing features, alerts, and incident information for operator review.

VIABILITY

Experimental evaluation: The tuned XGBoost model achieved 97.03% accuracy, 98.71% ransomware recall, and 99.54% ROC-AUC on the MLRan benchmark test set.

Live monitoring demo: The agent collects file-system, process, CPU, memory, and system telemetry. The dashboard displays events, alerts, incidents, and risk information through the backend API.

Security controls: Canary files, heartbeat monitoring, mass-change detection, rotating files, alert reporting, and automated isolation simulations support operator oversight.

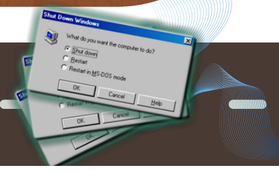
Prototype decision support: Displays behavioral detections, model predictions, 0–100 risk scores, threat explanations, alert status, and simulated response actions for investigation.

CHALLENGES AND RISKS

Data limitations: Benchmark datasets may not represent real institutional activity; workload variation, server-log imbalance, and unfamiliar ransomware behavior may affect detection accuracy.

Planned mitigation: Test with representative institutional workloads, review false positives, validate the 0.40 ML threshold, and tune security rules and alert limits.

Open risks: Detection of novel ransomware variants, false positives, production throughput, concept drift, system resource usage, and ongoing model evaluation still need testing.



IMPACT AND BENEFITS

Cipherveil Detection Workflow

Tracks device events, file activity, model predictions, system telemetry, and alert outcomes.

Monitors Device Activity

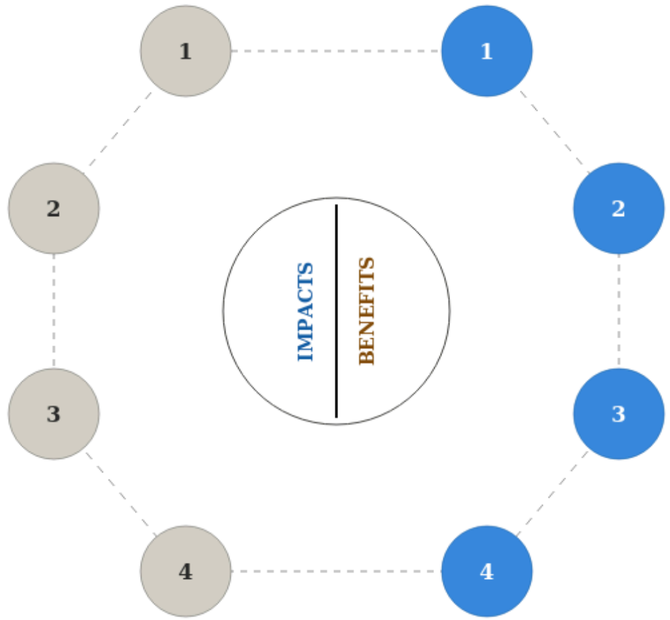
The monitoring agent collects file-system events, process activity, system resources, and device telemetry.

Detects Suspicious Behavior

The ML model analyzes behavioral feature windows and assigns a calibrated ransomware risk score.

Earlier Threat Response

Generates alerts, records incident information, calculates risk levels, and supports simulated containment through the security module.



Explainable Security Decisions

SHAP feature contributions identify important indicators. Alerts also include the prediction, risk score, and threshold result.

Layered Detection

Combines behavioral ML prediction, TreeSHAP explainability, canary checks, heartbeat monitoring, and mass-change detection.

Automated Threat Analysis

The ML engine processes telemetry, applies the tuned 0.40 decision threshold, and sends detection results to the backend.

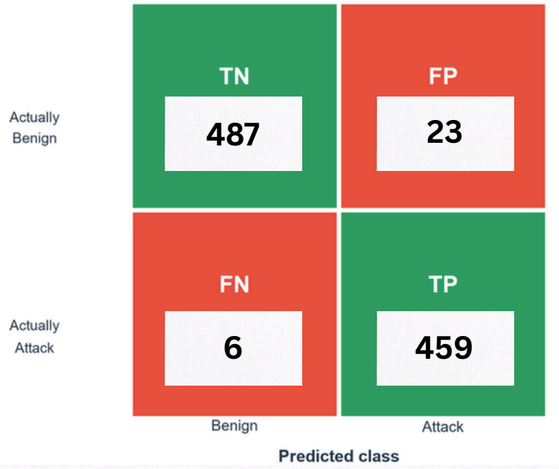
Benchmark Evaluation

Evaluates the XGBoost classifier on the MLRan test split and measures accuracy, recall, precision, false-positive rate, and ROC-AUC.

PROTOTYPE RESULTS

Metric	MLRan XGBoost classifier
Precision	95.23%
Recall	98.71%
F1-score	96.94%
Accuracy	97.03%

MLRan Ransomware Classifier: Tuned XGBoost



RESEARCH AND REFERENCES

Defines the Cipherveil architecture, including monitoring, ML prediction, security rules, backend services, and dashboard workflows.

Cipherveil System Architecture

LINK

Highlights security-layer controls such as rotating canary files, heartbeat monitoring, mass-change detection, API authentication, and response logging.

OWASP Top 10 for API Security

LINK

MLRan behavioral dataset for evaluating XGBoost-based ransomware classification on endpoint features.

MLRan Behavioral Dataset

LINK

Evaluates ML and security risks, including false positives, false negatives, threshold tuning, model drift, adversarial ransomware behavior, and explainability.

OWASP Top 10 for Machine Learning Security

LINK

Serves as a reference for layered endpoint risk analysis, combining ML predictions, security rules, alert evidence, and operator review.

SHAP Explainability Framework

LINK

Supports safe prototype evaluation using the monitoring simulator, benign test activity, canary simulation, heartbeat-loss testing, and containment-state simulation.

Safe Ransomware Simulator

LINK